

# Ideation Canvas

## Two Project Proposals

### Project 1: Aurora

Platform: Android

An Android app that detects hidden spyware on a phone, helps the user remove it safely without alerting the person who installed it, preserves the evidence, and provides a personalized digital safety checklist — all disguised as a weather app.

#### The problem

Stalker ware is software secretly installed on someone's phone — usually by a partner, ex-partner, or stalker — that silently transmits the victim's location, messages, calls, and photos to the person watching them. It does not show up in the normal app list. It does not announce itself. And removing it without a plan can be more dangerous than leaving it in place, because the abuser will immediately know it was found.

This is a documented, growing problem. A 2024 Kaspersky report found that over 31,000 people worldwide discovered Stalker ware on their phones in 2023, up 6% from the year before. A separate global survey found that 40% of people have experienced stalking or suspected they were being stalked. The problem is concentrated on Android — Stalker ware on iOS requires either a jailbroken device or a corporate MDM profile, making Android the platform where the vast majority of cases occur.

Domestic violence organizations, the Electronic Frontier Foundation, and the Coalition Against Stalker ware have all publicly identified the same tooling gap: victims need something that actively finds the spyware, explains what it is in plain language, helps them decide what to do about it safely, and preserves the evidence. That tool does not exist yet.

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| <b>31,031</b><br>people found Stalker ware on their phones in 2023 — up 6% year over year | <b>40%</b><br>of people globally have experienced or suspected stalking | <b>195</b><br>unique Stalker ware apps detected in 2023 — most sold openly as parental control tools |
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## Why existing tools do not solve this

Two categories of tools exist today, and both fall short for the people who need help most. Domestic violence support apps — including Better Days, built by a MICS team in Fall 2021 — provide resources, guides, and safety information. They do not scan for Stalker ware, do not remove it, and do not preserve evidence. Better Days' own threat model presentation identified spyware as an unresolved gap in their product.

Antivirus apps technically detect some Stalker ware, but are not built for victims. A 2025 joint test by AV-Comparatives and the Electronic Frontier Foundation found that most antivirus apps display only a generic alert with no context, no guidance, and no consideration of the danger that removal itself can create. The EFF's own guidance states clearly: removing Stalker ware without a safety plan can put the victim in greater danger. Aurora is built specifically for this situation — someone who needs to find out what is on their phone, understand what it means, and take action without making things worse.

## Who uses Aurora

|                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                           |
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| <b>People being monitored without their knowledge</b><br>Domestic violence victims, stalking victims, and anyone in a controlling relationship who suspects their phone is being used against them. This group cannot simply delete a suspicious app — doing so without a plan can alert the abuser and create real danger. | <b>Domestic violence advocates and shelter workers</b><br>Professionals who scan phones on behalf of clients at shelters and legal aid organizations. They need exportable, formatted evidence reports that can be submitted to law enforcement or used in restraining order proceedings. |
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## MVP scope — what we will build this semester

Scope discipline is critical with four people and one semester. Aurora's MVP is four features on Android — each one complete and demo-able, each one addressing a specific gap no existing tool fills. iOS support, AI-generated safety plans, and a multi-user advocate dashboard are explicitly out of scope for this semester.

### IN SCOPE FOR MVP

Feature 1: Android Stalker ware scanner using the Coalition Against Stalker ware database plus behavioral detection

Feature 2: Trauma-informed safe removal decision tree — three options based on the user's safety situation

Feature 3: Cryptographic evidence vault with PDF export for legal proceedings

Feature 4: Cover weather app with panic button

### OUT OF SCOPE — FUTURE PHASES

iOS support — Android covers the vast majority of Stalker ware cases; iOS requires a full separate build

AI-generated personalized safety plan — replaced with a structured checklist for the MVP

Multi-user advocate dashboard — single user flow only for this semester

Remote evidence wipe — complex infrastructure, post-MVP

## WHAT AURORA DOES — FOUR FEATURES

### 1. DETECT THE STALKER WARE

Aurora scans every installed app on the Android device using two methods. First, it checks against the Coalition Against Stalker ware's open-source database of known Stalker ware signatures — the established standard maintained by cybersecurity companies, NGOs, and law enforcement. Second, a behavioral layer analyzes suspicious patterns that signature scanning alone misses: apps combining location and microphone access with no legitimate purpose, apps transmitting large volumes

of data to unknown servers in the background, apps hidden from the standard app list, and abuse of Android Accessibility Services — the most common way Stalker ware evades detection. Results are displayed in plain language: not a generic malware alert, but a specific description of what the app is doing and to whom it is sending data.

## **2. REMOVE IT SAFELY**

Before any action is taken, Aurora asks three plain-language questions: are you still in contact with this person, do they have regular physical access to your phone, and are you currently in a safe location. The answers determine which of three paths is offered. If the abuser could notice the removal, Aurora provides a guided removal with a plausible explanation for why the app disappeared, or a scheduled removal option for when the user reaches safety. If the user is already out of the situation, it walks through full removal step by step. Nothing is ever deleted automatically — the Electronic Frontier Foundation and the Coalition Against Stalker ware both state that informed choice is essential, and Aurora is built around that principle.

## **3. PRESERVE THE EVIDENCE**

At the moment stalker ware is detected, Aurora creates a tamper-evident record. It generates a cryptographic hash of the scan results — a mathematical fingerprint proving the evidence existed at that exact time and has not been modified since. Everything is stored in an AES-256 encrypted vault on the device, presented to the user as a photo album, so it does not attract attention. At any point, the user can generate a formatted PDF report showing the app name, installation indicators, permissions granted, data transmission behavior, and the cryptographic proof of discovery. This report is formatted for submission to law enforcement or use in a restraining order application.

## **4. COVER APP AND PANIC BUTTON**

Aurora opens as a fully functional weather app — real data, real forecasts, nothing suspicious. A specific hidden gesture launches the actual tool. A panic button on any screen within the real app returns instantly to the weather view and clears recent activity if someone picks up the phone mid-use. The app is listed in the device's app drawer under a neutral name. The name Aurora returns no security-related results in

a search — the northern lights, a Colorado city, a Disney character — nothing that would raise concern. The cover is a security control, not a cosmetic choice.

## Threat model

Aurora's threat model centers on a specific adversary: the person who installed the Stalker ware and who may have ongoing access to the victim's device, network traffic, or physical environment. Every design decision in Aurora maps to a documented threat from this adversary.

| Threat                       | Attack scenario                              | Aurora's mitigation                                                                     |
|------------------------------|----------------------------------------------|-----------------------------------------------------------------------------------------|
| Abuser discovers Aurora      | Opens the app on the victim's phone          | Fully functional weather cover app, hidden gesture to enter real tool                   |
| Abuser discovers Aurora      | Googles the app name                         | Name returns only weather, geography, and entertainment results                         |
| Abuser discovers Aurora      | Monitors network traffic during a scan       | CAS database updates are timed and disguised — no suspicious traffic spike during scan  |
| Removal triggers retaliation | Abuser notices Stalker ware has been removed | Three-path removal: coached explanation, scheduled for safety, or document-and-wait     |
| Evidence vault is found      | Abuser seizes the phone and searches it      | Vault appears as a photo album, AES-256 encrypted, decoy content visible                |
| Aurora leaks user data       | App sends victim data to external servers    | Zero telemetry, no account required, all processing on-device, no cloud sync by default |

## Evaluation criteria

The following maps Aurora directly to the MICS capstone judging guidelines.

### Originality

No existing tool combines Stalker ware detection, trauma-informed removal guidance, cryptographic evidence preservation, and a functioning cover app in a single product. Better Days (MICS Fall 2021) addressed the same population but built a resource library

rather than a detection tool — their threat model slide explicitly identified Stalker ware as an unresolved gap. Aurora builds what they identified but could not complete.

### Potential impact and meeting a need

Over 31,000 people discovered Stalker ware on their devices in 2023. The EFF, the Coalition Against Stalker ware, and NNEDV's Safety Net Project have all publicly described this tooling gap. Domestic violence advocates currently have no automated way to scan client devices — they rely on manual inspection or generic antivirus tools that were not designed for this situation. Aurora addresses a specific, documented, and currently unmet need.

### Completeness

Four features, Android only, four people, one semester. The scope is deliberately constrained so that every feature ships working and demoable. The MVP boundary is defined in this document: what is in, what is explicitly out, and why. A well-scoped project that does four things completely is more valuable than an ambitious project that does eight things partially.

### Security thinking

The threat model is the product. Every Aurora feature is a direct response to a specific adversary behavior from the person who installed the Stalker ware. The technical architecture reflects this: AES-256 evidence encryption, SHA-256 tamper-evident hashing of scan artifacts, zero telemetry, all processing on-device, behavioral detection for evasive Stalker ware that hides from signature scanning, and a three-path removal decision tree designed to prevent the victim from triggering retaliation. Secure by design is not a layer added on top of the product — it is the reason the product is designed the way it is.

### Presentation

The demo is concrete and experiential: open a weather app, use a hidden gesture to enter the real tool, run a scan on a pre-loaded Android device, find Stalker ware in under 30 seconds, read the plain-language alert, walk through the removal options, view the evidence vault with its cryptographic proof, press the panic button and return to the weather screen. Every person in the room will understand immediately what the tool does and why it matters — no technical background required.

## Technical foundation

- Coalition Against Stalker ware open-source signature database — actively maintained detection backbone
- Android PackageManager API — installed app enumeration and permission auditing
- Accessibility Service enumeration — detecting the most common Stalker ware evasion technique
- Device Administrator API — identifying apps with elevated device control
- Background network traffic analysis — detecting covert data exfiltration patterns
- AES-256 local encryption for the evidence vault
- SHA-256 hashing for tamper-evident discovery records
- PDF report generation for law enforcement and legal proceedings

## Key sources and potential partners

- Coalition Against Stalker ware ([stopStalkerware.org](https://stopStalkerware.org)) — open-source detection database, best practices, and advisory potential
- National Network to End Domestic Violence, Safety Net Project — gold standard for technology safety planning in DV contexts
- Electronic Frontier Foundation — Stalker ware detection research and removal guidelines
- Kaspersky State of Stalker ware 2023 Report — primary data source for scale of the problem
- AV-Comparatives + EFF Stalker ware Detection Test 2025 — technical analysis of existing tool gaps

# Project 2: Compass

Platform: Web application

A web tool that scans how any organization is using AI tools like ChatGPT or Microsoft Copilot, identifies their governance and security gaps against the NIST AI Risk Management Framework, and delivers the findings in plain language to three different audiences: the CEO, the security team, and the auditor.

## The problem

AI adoption has moved faster than the governance structures around it. In 2024, the amount of corporate data flowing into AI tools like ChatGPT increased 485% compared to the year before — and nearly a quarter of that data was sensitive: customer records, financial information, proprietary code, personal health details. Most organizations deploying these tools have no policy governing their use, no oversight process, and no systematic way to assess their exposure.

The National Institute of Standards and Technology published its AI Risk Management Framework in 2023 and expanded it specifically for generative AI tools in July 2024. It is the most authoritative guidance available for how organizations should manage AI risk. The gap between the framework existing and organizations actually using it is significant: 77% of organizations say they are building AI governance programs, but only 36% have adopted a formal framework. That gap has real financial consequences — a 2025 EY survey of nearly a thousand C-suite leaders found that 99% of organizations reported financial losses from AI-related risks, with nearly two-thirds losing more than one million dollars.

The reason most organizations have not adopted the framework is not lack of interest — it is that the NIST AI RMF is a 70-page document written in technical framework language. No automated tool exists that translates it into a practical assessment any organization can complete in under an hour. Compass builds that tool.

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| <b>485%</b><br>increase in corporate data<br>entering AI tools between 2023<br>and 2024 | <b>77% vs 36%</b><br>organizations wanting AI<br>governance vs. those with a<br>real framework in place | <b>99%</b><br>of C-suite leaders reported<br>financial losses from AI-related<br>risk in 2025 |
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## Why existing tools do not solve this

Enterprise AI governance platforms like ServiceNow AI Governance or IBM OpenPages are designed for large organizations with dedicated compliance teams, significant budgets, and months to implement. They are not built for the company whose employees started using ChatGPT last quarter and whose security lead now needs to explain the risk posture to the board.

Manual consulting assessments cost thousands of dollars, take weeks, and produce a PDF that is out of date by delivery. Spreadsheet-based self-assessments require someone to read the full framework document and score themselves against it — something almost no one completes correctly or consistently.

Compass sits between those two options: automated, fast, and designed for organizations that are not framework experts. Rather than presenting the NIST AI RMF as a document to read, Compass runs an AI-guided conversation that asks plain-language questions and maps the answers to specific framework controls automatically.

## Who uses Compass

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| <b>Any organization using AI tools</b><br>Companies using ChatGPT, Microsoft Copilot, Google Gemini, or any AI product that touches company data. The primary user is the person who has just been asked by their board or a regulator to describe their AI governance program and does not yet have a structured answer. | <b>Security leads and compliance officers</b><br>People responsible for reporting AI risk posture to boards and regulators, currently doing it manually via spreadsheets or expensive external consultants. Compass gives them an automated, repeatable process they can re-run as their AI footprint changes. |
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# What Compass does — four features

## 1. Build an AI inventory

Organizations often do not have a complete picture of which AI tools their employees are actually using. Departments adopt tools independently, individuals bring in their own accounts, and no one maintains a master list. Compass starts by building that inventory through a structured intake — asking about tools in use, how they connect to company systems, and what categories of data they process. This maps to the NIST AI RMF's MAP function, which requires organizations to identify and understand their AI systems before they can manage the risks those systems create.

## 2. Run a plain-language gap assessment

Rather than asking users to read a 70-page framework document, Compass conducts an AI-guided conversation. It asks plain-language questions — Do you have a written policy for which AI tools employees are allowed to use? If your AI tool produces a wrong answer that affects a customer, who in your organization is responsible? — and maps the answers automatically to the four functions of the NIST AI RMF: Govern, Map, Measure, and Manage. The assessment also covers the 12 generative AI-specific risks NIST identified in its July 2024 update: hallucination, data poisoning, prompt injection, intellectual property exposure, and others.

## 3. Deliver findings to three audiences from one scan

The same assessment produces three views. The executive view translates gaps into business consequences: your employees are sending customer data to an external AI service with no contractual protections in place — this creates legal and financial exposure. The security team view shows specific control gaps mapped to NIST framework functions, with a risk score and priority ranking for each. The auditor or regulator view produces a formatted compliance report with evidence of the assessment process. One tool, one scan, three audiences — each getting the level of detail they can actually act on.

## 4. Generate a prioritized action plan

A gap assessment is only useful if it produces action. Compass generates a remediation roadmap showing which gaps to address first, what policies to create,

which NIST AI RMF playbook actions apply to each finding, and how to measure improvement over time. Organizations can re-run the assessment after making changes. The final output is a formatted PDF report suitable for a board presentation or a regulatory submission.

## Evaluation criteria

The following maps Compass directly to the MICS capstone judging guidelines.

### Originality

No tool currently translates the NIST AI RMF — including the July 2024 GenAI Profile covering risks specific to tools like ChatGPT — into an automated, plain-language assessment accessible to any organization without a compliance team. Enterprise platforms exist but are priced and scoped for large organizations. Compass serves the organizations that those platforms ignore.

### Potential impact and meeting a need

77% of organizations are building AI governance programs but only 36% have a real framework. 99% are losing money to AI risk. Every company using ChatGPT, Copilot, or any AI product touching sensitive data is a potential user. The 2025 Pacific AI Governance Survey found that fewer than 20% of organizations have AI incident reporting tools in place even in heavily regulated industries — Compass addresses a gap that is both large and actively growing.

### Completeness

Four features, web application, four people, one semester. AI inventory intake, NIST-mapped questionnaire engine, three-audience output dashboard, and remediation roadmap. All four are demoable as a working product using a fictional company scenario. Scope is focused and achievable within the semester.

### Security thinking

Compass is itself a security product — it identifies AI security and governance gaps in real organizations and maps them to the NIST AI RMF. The knowledge graph encodes the full framework including the GenAI Profile's 12 risk categories: hallucination, data poisoning, prompt injection, IP exposure, over-reliance, and more. Compass must also protect the organizational data that users share during the assessment: intake responses are

processed in-session and not retained after the user exports their report. The system applies the same security principles it is evaluating.

## Presentation

The demo is concrete: enter a fictional company profile — uses ChatGPT for customer support, no written AI policy, no oversight process. Run the assessment. Watch the gap dashboard populate: Govern failing, Map partial, Measure absent, Manage absent. Executive view: your customer support team is sending complaints including names and account numbers to an external AI service with no data agreement in place. Security view: specific gaps listed with priority scores. Auditor view: formatted PDF. Remediation roadmap generated. Total demo time: under two minutes.

## Technical foundation

- NIST AI RMF 1.0 (January 2023) encoded as a structured knowledge graph — all controls, functions, and sub-categories
- NIST-AI-600-1 GenAI Profile (July 2024) — 12 generative AI-specific risks integrated into the assessment
- LLM-powered questionnaire engine — adapts follow-up questions based on prior answers, generates plain-language explanations of each gap
- Gap scoring algorithm — weights findings by risk severity and likelihood of harm, produces priority-ranked output
- Three-audience rendering layer — same underlying gap data formatted differently for executive, security, and auditor outputs
- Secure session handling — intake responses processed in memory, not persisted after report export

## Key sources

- NIST AI Risk Management Framework 1.0 (January 2023) and GenAI Profile NIST-AI-600-1 (July 2024)
- 2025 IAPP AI Governance Profession Report — 77% building governance, 36% with a formal framework
- 2025 EY Global C-Suite AI Risk Survey — 99% of leaders reporting financial losses

- 2025 Pacific AI Governance Survey — fewer than 20% of orgs have AI incident reporting tools
- Wiz State of AI Security 2025 — 485% increase in corporate data in AI tools

## Side by side

|                               | Aurora                                                                                                    | Compass                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Platform                      | Android                                                                                                   | Web application                                                                                         |
| Primary audience              | Stalking and domestic violence victims, shelter advocates                                                 | Any organization using ChatGPT, Copilot, or similar AI tools                                            |
| The core problem              | 31,000+ people have hidden spyware on their phones with no safe tool to find or remove it                 | 77% of organizations want AI governance — only 36% have built one — and 99% are losing money to AI risk |
| Originality                   | Never built in 20 semesters of MICS capstones                                                             | Never built in 20 semesters of MICS capstones                                                           |
| MVP achievable this semester? | Yes — 4 features, Android only, well-defined scope boundary                                               | Yes — 4 features, web app, well-defined scope boundary                                                  |
| Security thinking criterion   | Strongest possible — threat model is the entire product architecture                                      | Strong — NIST knowledge graph, 12 GenAI risk categories, secure session handling                        |
| Demo                          | Live scan finds Stalker ware in 30 seconds — judges watch it happen on a real device                      | Fictional company scenario shows gap dashboard populating in real time — under 2 minutes                |
| Technical complexity          | Behavioral detection engine, cryptographic evidence chain, trauma-informed decision tree, cover app opsec | NIST knowledge graph, LLM questionnaire engine, gap scoring algorithm, three-audience output layer      |
| Real-world partners available | Coalition Against Stalker ware, NNEDV Safety Net Project, EFF                                             | Any organization willing to pilot the assessment tool                                                   |